

Introduction to Approximate Inference

Advanced Statistical Inference

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Bayesian inference

Bayesian inference allows to “transform” a prior distribution over the parameters into a posterior **after** observing the data

Prior distribution $p(\mathbf{w})$

Posterior distribution $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$

Bayes’ rule:

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}$$

- **Prior:** $p(\mathbf{w})$
 - Encodes our beliefs about the parameters **before** observing the data
- **Likelihood:** $p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})$
 - Encodes our model of the data
- **Posterior:** $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$
 - Encodes our beliefs about the parameters **after** observing the data (e.g. conditioned on the data)
- **Evidence:** $p(\mathbf{y} \mid \mathbf{X})$
 - Normalizing constant, ensures that $\int p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) d\mathbf{w} = 1$

Bayesian linear regression (review)

Modeling observation as noisy realization of a linear combination of the features As before, we assume a Gaussian likelihood

$$p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}) = \mathcal{N}(\mathbf{y} \mid \mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I})$$

For the prior, we use a Gaussian distribution over the model parameters

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \mathbf{S})$$

In practice, we often use a diagonal covariance matrix $\mathbf{S} = \sigma_w^2 \mathbf{I}$

When can we compute the posterior?

Definition

A prior is **conjugate** to a likelihood if the posterior is in the same family as the prior.

Only a few conjugate priors exist, but they are very useful.

Examples:

- Gaussian likelihood and Gaussian prior $\Rightarrow$ Gaussian posterior
- Binomial likelihood and Beta prior $\Rightarrow$ Beta posterior

Full table available on [wikipedia](#)

Why is this useful?

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}$$

- **Generally** the posterior is **intractable** to compute
 - We don't the form of the posterior $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$
 - The evidence $p(\mathbf{y} \mid \mathbf{X})$ is an integral
 - * $p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w}) d\mathbf{w}$
 - * without closed form solution
 - * high-dimensional and computationally intractable to compute numerically

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- **Analytical solution** thanks to conjugacy:
 - We know the form of the posterior
 - We know the form of the normalization constant
 - We don't need to compute the evidence, just some algebra to get the posterior
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From the likelihood and prior, we can write the posterior as

$$\begin{aligned}
 p(\mathbf{y} \mid \mathbf{w}, \mathbf{X})p(\mathbf{w}) &\propto \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 - \frac{1}{2}\mathbf{w}^\top \mathbf{S}^{-1}\mathbf{w}\right) \\
 &\propto \exp\left(-\frac{1}{2}\left(\mathbf{w}^\top \left(\frac{1}{\sigma^2}\mathbf{X}^\top \mathbf{X} + \mathbf{S}^{-1}\right)\mathbf{w} - \frac{2}{\sigma^2}\mathbf{w}^\top \mathbf{X}^\top \mathbf{y}\right)\right)
 \end{aligned}$$

...

From conjugacy, we know that the posterior is Gaussian

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X}) \propto \exp\left(-\frac{1}{2}\left(\mathbf{w}^\top \boldsymbol{\Sigma}^{-1}\mathbf{w} - 2\mathbf{w}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}\right)\right)$$

We can identify the posterior mean and covariance

Posterior covariance

$$\boldsymbol{\Sigma} = \left(\frac{1}{\sigma^2}\mathbf{X}^\top \mathbf{X} + \mathbf{S}^{-1}\right)^{-1}$$

Posterior mean

$$\boldsymbol{\mu} = \frac{1}{\sigma^2}\boldsymbol{\Sigma}\mathbf{X}^\top \mathbf{y}$$

Exact inference is rare

- **Exact inference** is possible when the posterior distribution can be computed analytically

Example: Linear regression with Gaussian likelihood and Gaussian prior

- This is the case for simple models with conjugate priors, ...
- ... but most of the time, the posterior distribution is intractable

Examples: Logistic regression (binary classification), neural networks, ...

Introduction to Approximate Inference

- In this lecture, we will introduce the concept of **approximate inference** in the context of Bayesian models.
- Approximate inference methods provide a way to approximate the posterior distribution when it is intractable
- For the next 2 weeks, we will be **model-agnostic** and focus on the methods used to perform inference in complex and intractable models.

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Why model-agnostic?

Solving a machine learning problem involves multiple steps:

1. **Modeling:** Define a model that captures the underlying structure of the data
2. **Inference:** Estimate the parameters of the model
3. **Prediction:** Use the model to make predictions on new data

In these two weeks, we will focus on the **inference** step

Problem definition

In probabilistic models, all unknown quantities are treated as **random variables**

- **Observed quantities:** $\mathbf{y} \in \mathbb{R}^N$ (vector of N observations);
- **Unknown variables:** $\boldsymbol{\theta} \in \mathbb{R}^D$ (vector of D parameters)

Given a likelihood $p(\mathbf{y} \mid \boldsymbol{\theta})$ and a prior $p(\boldsymbol{\theta})$, the goal is to compute the posterior distribution $p(\boldsymbol{\theta} \mid \mathbf{y})$

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})}$$

Note: We drop the conditioning on the data $\mathbf{X}$ for simplicity, but it is present in the likelihood as input of the model

Approximate inference

Approximate inference methods provide a way to approximate distributions when the exact computation is intractable

$$p(\boldsymbol{\theta} \mid \mathbf{y}) \approx q(\boldsymbol{\theta})$$

We will study two main classes of approximate inference methods



Sampling-based methods

- **Monte Carlo methods**
- **Markov Chain Monte Carlo (MCMC)**
- **Hamiltonian Monte Carlo (HMC)**



Parametric methods

- **Variational inference**
- **Laplace approximation**

Grid approximation

Divide the parameter space into K regions $\mathcal{R}_1, \dots, \mathcal{R}_K$ of equal volume Δ . For each region, approximate the posterior mass by a Riemann approximation:

$$p(\boldsymbol{\theta} \in \mathcal{R}_k \mid \mathbf{y}) = \int_{\mathcal{R}_k} p(\boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta} \approx p(\boldsymbol{\theta}_k \mid \mathbf{y}) \Delta.$$

where $\boldsymbol{\theta}_k$ is a representative point of the region $\mathcal{R}_k$ (e.g. the center of the region). Use Bayes' rule at each grid point:

$$p(\boldsymbol{\theta}_k \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}_k)p(\boldsymbol{\theta}_k)}{\int p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta}) d\boldsymbol{\theta}}.$$

Define unnormalized terms $\tilde{p}_k = p(\mathbf{y} \mid \boldsymbol{\theta}_k)p(\boldsymbol{\theta}_k)$. Normalization gives

$$p(\boldsymbol{\theta}_k \mid \mathbf{y}) = \frac{\tilde{p}_k}{\sum_{j=1}^K \tilde{p}_j}, \quad p(\boldsymbol{\theta} \in \mathcal{R}_k \mid \mathbf{y}) \approx \frac{\tilde{p}_k}{\sum_{j=1}^K \tilde{p}_j} \Delta.$$

The marginal likelihood follows from the same Riemann sum $p(\mathbf{y}) \approx \sum_{k=1}^K \tilde{p}_k \Delta$

